2 PART

```
olya@firmware-vm:~/extraction_07/_C9200_9300_9400_9500_9600_cat9k_iosxe.16.12.07.SPA.bin.extracted/squashfs-root$ ls -lh
total 836M
-rw-r--r-- 1 olya olya  20M лют  2  2022 cat9k-cc_srdriver.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya  85M лют  2  2022 cat9k-espbase.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya 1,7M лют  2  2022 cat9k-guestshell.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya 512M лют  2  2022 cat9k-rpbase.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya  30M лют  2  2022 cat9k-sipbase.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya  58M лют  2  2022 cat9k-sipspa.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya  30M лют  2  2022 cat9k-srdriver.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya  13M лют  2  2022 cat9k-webui.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya 9,1K лют  2  2022 cat9k-wlc.16.12.07.SPA.pkg
-rw-r--r-- 1 olya olya 6,5K бер  2 18:16 packages_07.conf
-rw-r--r-- 1 olya olya 6,5K лют  2  2022 packages.conf
-rw-r--r-- 1 olya olya 1,7K бер  2 20:14 pkgscan.go
-rw-rw-r-- 1 olya olya  89M бер  2 20:28 rpbase_payload.bin
olya@firmware-vm:~/extraction_07/_C9200_9300_9400_9500_9600_cat9k_iosxe.16.12.07.SPA.bin.extracted/squashfs-root$ _
```

Among the extracted IOS-XE packages, the file cat9k-rpbase.16.12.07.SPA.pkg has the largest size (~512 MB). Because of this, it most likely contains the core runtime components of the system.
For this reason, further structural analysis will focus primarily on this package. Analyzing the largest package increases the chance of discovering the internal container structure, metadata layout, and embedded payload segments used by the firmware.

Given that the file size is approximately 512 MB, this binary payload likely stores compressed modules or filesystem components used by the IOS-XE runtime environment.

MetaData

cat9k-rpbase.16.12.0...

```
00000000  40 D1 81 FF 78 D5 46 8E A3 B6 37 0B 8D CA B7 DC  @╤ü x╒FÄú╢7.ì╩╖▄
00000010  82 24 21 3D A9 5B 37 B1 9B 34 EF BD 76 03 22 53  é$!=⌐[7▒¢4∩╝v."S
00000020  E1 B3 7F 66 E5 CF C3 96 D2 CB E2 F0 D0 BF 40 7D  ß│⌂fσ╧├û╥╦╥≡╨┐@}
00000030  30 E1 C8 DA 00 00 00 00 DE AD BE EF DE AD BE EF  0ß╚┌....▌¡╛∩▌¡╛∩
00000040  00 00 04 04 00 00 75 32 00 00 00 00 00 00 04 04  ......u2........
00000050  1F FE 14 04 00 00 00 00 00 00 00 00 00 00 00 00  .▪..............
00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000070  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000080  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 03  ................
00000090  00 00 00 09 00 00 00 1D 4B 45 59 5F 54 4C 56 5F  ........KEY_TLV_
000000A0  50 41 43 4B 41 47 45 5F 43 4F 4D 50 41 54 49 42  PACKAGE_COMPATIB
000000B0  49 4C 49 54 59 00 00 00 00 00 00 09 00 00 00 0B  ILITY...........
000000C0  46 52 55 5F 52 50 5F 54 59 50 45 00 00 00 00 09  FRU_RP_TYPE.....
000000D0  00 00 00 18 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000000E0  41 47 45 5F 42 4F 4F 54 41 52 43 48 00 00 00 09  AGE_BOOTARCH....
000000F0  00 00 00 0E 41 52 43 48 5F 69 36 38 36 5F 54 59  ....ARCH_i686_TY
00000100  50 45 00 00 00 00 00 09 00 00 00 14 4B 45 59 5F  PE..........KEY_
00000110  54 4C 56 5F 42 4F 41 52 44 5F 43 4F 4D 50 41 54  TLV_BOARD_COMPAT
00000120  00 00 00 09 00 00 00 10 42 4F 41 52 44 5F 63 61  ........BOARD_ca
00000130  74 39 6B 5F 54 59 50 45 00 00 00 09 00 00 00 18  t9k_TYPE........
00000140  4B 45 59 5F 54 4C 56 5F 43 52 59 50 54 4F 5F 4B  KEY_TLV_CRYPTO_K
00000150  45 59 53 54 52 49 4E 47 00 00 00 09 00 00 00 04  EYSTRING........
00000160  6E 6F 6E 65 00 00 00 09 00 00 00 0B 43 57 5F 42  none........CW_B
00000170  45 47 49 4E 3D 24 24 00 00 00 00 09 00 00 00 11  EGIN=$$.........
00000180  43 57 5F 46 41 4D 49 4C 59 3D 24 63 61 74 39 6B  CW_FAMILY=$cat9k
00000190  24 00 00 00 00 00 00 09 00 00 00 28 43 57 5F 49  $..........(CW_I
000001A0  4D 41 47 45 3D 24 63 61 74 39 6B 2D 72 70 62 61  MAGE=$cat9k-rpba
000001B0  73 65 2E 31 36 2E 31 32 2E 30 37 2E 53 50 41 2E  se.16.12.07.SPA.
000001C0  70 6B 67 24 00 00 00 09 00 00 00 15 43 57 5F 56  pkg$........CW_V
000001D0  45 52 53 49 4F 4E 3D 24 31 36 2E 31 32 2E 30 37  ERSION=$16.12.07
000001E0  24 00 00 00 00 00 00 09 00 00 00 37 43 57 5F 46  $..........7CW_F
000001F0  55 4C 4C 5F 56 45 52 53 49 4F 4E 3D 24 31 36 2E  ULL_VERSION=$16.
00000200  31 32 2E 30 37 2E 30 2E 36 35 36 35 2E 31 36 34  12.07.0.6565.164
00000210  33 38 32 39 39 39 36 2E 2E 47 69 62 72 61 6C 74  3829996..Gibralt
00000220  61 72 24 00 00 00 00 09 00 00 00 34 43 57 5F 44  ar$........4CW_D
00000230  45 53 43 52 49 50 54 49 4F 4E 3D 24 43 69 73 63  ESCRIPTION=$Cisc
```

For the initial structural inspection of the .pkg container, the analysis was planned to focus on the first 4–8 KB of the file, which corresponds to offsets 0x00001000 (4096 bytes) to 0x00002000 (8192 bytes) from the beginning of the package.

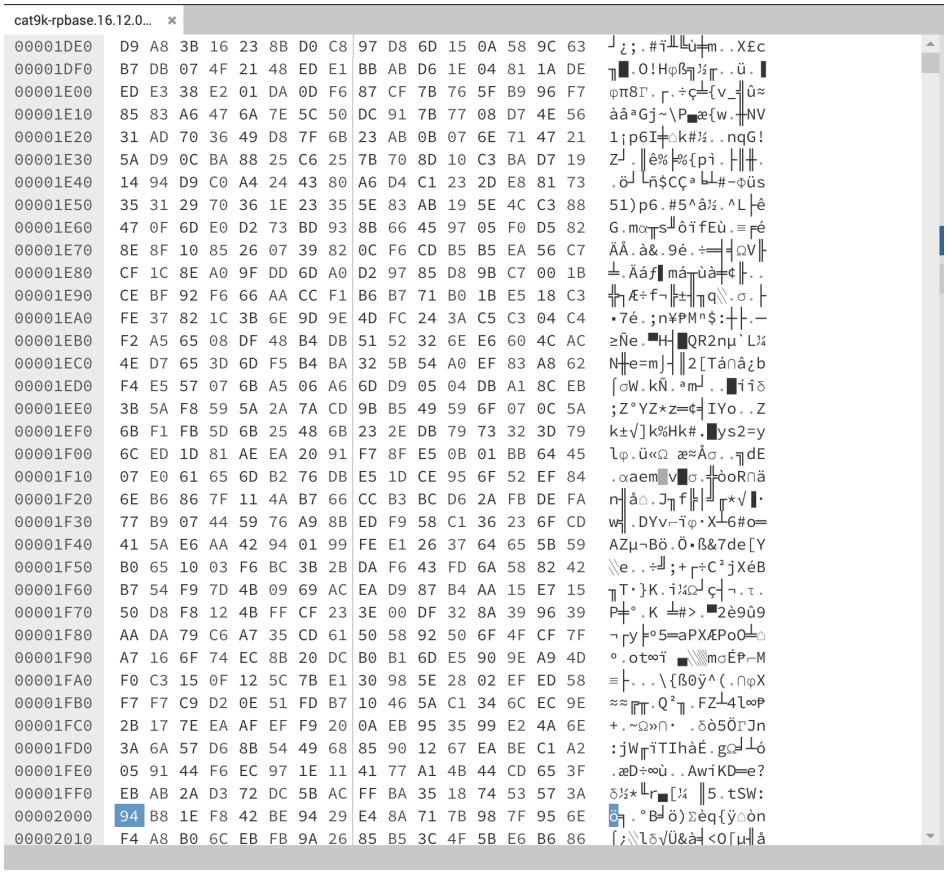
```
cat9k-rpbase.16.12.0...
00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000070  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000080  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 03  ................
00000090  00 00 00 09 00 00 00 1D 4B 45 59 5F 54 4C 56 5F  ........KEY_TLV_
000000A0  50 41 43 4B 41 47 45 5F 43 4F 4D 50 41 54 49 42  PACKAGE_COMPATIB
000000B0  49 4C 49 54 59 00 00 00 00 00 00 09 00 00 00 0B  ILITY...........
000000C0  46 52 55 5F 52 50 5F 54 59 50 45 00 00 00 00 09  FRU_RP_TYPE.....
000000D0  00 00 00 18 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000000E0  41 47 45 5F 42 4F 4F 54 41 52 43 48 00 00 00 09  AGE_BOOTARCH....
000000F0  00 00 00 0E 41 52 43 48 5F 69 36 38 36 5F 54 59  ....ARCH_i686_TY
00000100  50 45 00 00 00 00 00 09 00 00 00 14 4B 45 59 5F  PE..........KEY_
00000110  54 4C 56 5F 42 4F 41 52 44 5F 43 4F 4D 50 41 54  TLV_BOARD_COMPAT
00000120  00 00 00 09 00 00 00 10 42 4F 41 52 44 5F 63 61  ........BOARD_ca
00000130  74 39 6B 5F 54 59 50 45 00 00 00 09 00 00 00 18  t9k_TYPE........
00000140  4B 45 59 5F 54 4C 56 5F 43 52 59 50 54 4F 5F 4B  KEY_TLV_CRYPTO_K
00000150  45 59 53 54 52 49 4E 47 00 00 00 09 00 00 00 04  EYSTRING........
00000160  6E 6F 6E 65 00 00 00 09 00 00 00 0B 43 57 5F 42  none........CW_B
00000170  45 47 49 4E 3D 24 24 00 00 00 00 09 00 00 00 11  EGIN=$$.........
00000180  43 57 5F 46 41 4D 49 4C 59 3D 24 63 61 74 39 6B  CW_FAMILY=$cat9k
00000190  24 00 00 00 00 00 00 09 00 00 00 28 43 57 5F 49  $..........(CW_I
000001A0  4D 41 47 45 3D 24 63 61 74 39 6B 2D 72 70 62 61  MAGE=$cat9k-rpba
000001B0  73 65 2E 31 36 2E 31 32 2E 30 37 2E 53 50 41 2E  se.16.12.07.SPA.
000001C0  70 6B 67 24 00 00 00 09 00 00 00 15 43 57 5F 56  pkg$........CW_V
000001D0  45 52 53 49 4F 4E 3D 24 31 36 2E 31 32 2E 30 37  ERSION=$16.12.07
000001E0  24 00 00 00 00 00 00 09 00 00 00 37 43 57 5F 46  $..........7CW_F
000001F0  55 4C 4C 5F 56 45 52 53 49 4F 4E 3D 24 31 36 2E  ULL_VERSION=$16.
00000200  31 32 2E 30 37 2E 30 2E 36 35 36 35 2E 31 36 34  12.07.0.6565.164
00000210  33 38 32 39 39 39 36 2E 2E 47 69 62 72 61 6C 74  3829996..Gibralt
00000220  61 72 24 00 00 00 00 09 00 00 00 34 43 57 5F 44  ar$........4CW_D
00000230  45 53 43 52 49 50 54 49 4F 4E 3D 24 43 69 73 63  ESCRIPTION=$Cisc
00000240  6F 20 49 4F 53 20 53 6F 66 74 77 61 72 65 2C 20  o IOS Software,
00000250  49 4F 53 2D 58 45 20 53 6F 66 74 77 61 72 65 24  IOS-XE Software$
00000260  00 00 00 09 00 00 00 09 43 57 5F 45 4E 44 3D 24  ........CW_END=$
00000270  24 00 00 00 00 00 00 0C 00 00 01 88 01 00 02 01  $..........ê....
00000280  01 02 00 04 00 00 01 7C 03 00 61 04 00 28 43 4E  .......|..a..(CN
00000290  3D 43 69 73 63 6F 53 79 73 74 65 6D 73 3B 4F 55  =CiscoSystems;OU
```

During hex-level inspection of the beginning of the file, readable metadata entries were observed, including markers such as CW_BEGIN=$ and CW_END=$. These entries appear to be part of a structured metadata header that uses a TLV-like format, where configuration and compatibility information about the package is stored.

```
cat9k-rpbase.16.12.0...
000000D0  00 00 00 18 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000000E0  41 47 45 5F 42 4F 4F 54 41 52 43 48 00 00 00 09  AGE_BOOTARCH....
000000F0  00 00 00 0E 41 52 43 48 5F 69 36 38 36 5F 54 59  ....ARCH_i686_TY
00000100  50 45 00 00 00 00 00 09 00 00 00 14 4B 45 59 5F  PE..........KEY_
00000110  54 4C 56 5F 42 4F 41 52 44 5F 43 4F 4D 50 41 54  TLV_BOARD_COMPAT
00000120  00 00 00 09 00 00 00 10 42 4F 41 52 44 5F 63 61  ........BOARD_ca
00000130  74 39 6B 5F 54 59 50 45 00 00 00 09 00 00 00 18  t9k_TYPE........
00000140  4B 45 59 5F 54 4C 56 5F 43 52 59 50 54 4F 5F 4B  KEY_TLV_CRYPTO_K
00000150  45 59 53 54 52 49 4E 47 00 00 00 09 00 00 00 04  EYSTRING........
00000160  6E 6F 6E 65 00 00 00 09 00 00 00 0B 43 57 5F 42  none........CW_B
00000170  45 47 49 4E 3D 24 24 00 00 00 00 09 00 00 00 11  EGIN=$$.........
00000180  43 57 5F 46 41 4D 49 4C 59 3D 24 63 61 74 39 6B  CW_FAMILY=$cat9k
00000190  24 00 00 00 00 00 00 09 00 00 00 28 43 57 5F 49  $..........(CW_I
000001A0  4D 41 47 45 3D 24 63 61 74 39 6B 2D 72 70 62 61  MAGE=$cat9k-rpba
000001B0  73 65 2E 31 36 2E 31 32 2E 30 37 2E 53 50 41 2E  se.16.12.07.SPA.
000001C0  70 6B 67 24 00 00 00 09 00 00 00 15 43 57 5F 56  pkg$........CW_V
000001D0  45 52 53 49 4F 4E 3D 24 31 36 2E 31 32 2E 30 37  ERSION=$16.12.07
000001E0  24 00 00 00 00 00 00 09 00 00 00 37 43 57 5F 46  $..........7CW_F
000001F0  55 4C 4C 5F 56 45 52 53 49 4F 4E 3D 24 31 36 2E  ULL_VERSION=$16.
00000200  31 32 2E 30 37 2E 30 2E 36 35 36 35 2E 31 36 34  12.07.0.6565.164
00000210  33 38 32 39 39 39 36 2E 2E 47 69 62 72 61 6C 74  3829996..Gibralt
00000220  61 72 24 00 00 00 00 09 00 00 00 34 43 57 5F 44  ar$........4CW_D
00000230  45 53 43 52 49 50 54 49 4F 4E 3D 24 43 69 73 63  ESCRIPTION=$Cisc
00000240  6F 20 49 4F 53 20 53 6F 66 74 77 61 72 65 2C 20  o IOS Software,
00000250  49 4F 53 2D 58 45 20 53 6F 66 74 77 61 72 65 24  IOS-XE Software$
00000260  00 00 00 09 00 00 00 09 43 57 5F 45 4E 44 3D 24  ........CW_END=$
00000270  24 00 00 00 00 00 00 0C 00 00 01 88 01 00 02 01  $..........ê....
00000280  01 02 00 04 00 00 01 7C 03 00 61 04 00 28 43 4E  .......|..a..(CN
00000290  3D 43 69 73 63 6F 53 79 73 74 65 6D 73 3B 4F 55  =CiscoSystems;OU
000002A0  3D 49 4F 53 2D 58 45 3B 4F 3D 43 69 73 63 6F 53  =IOS-XE;O=CiscoS
000002B0  79 73 74 65 6D 73 05 00 08 36 31 46 41 45 35 36  ystems...61FAE56
000002C0  30 06 00 28 43 4E 3D 43 69 73 63 6F 53 79 73 74  0..(CN=CiscoSyst
000002D0  65 6D 73 3B 4F 55 3D 49 4F 53 2D 58 45 3B 4F 3D  ems;OU=IOS-XE;O=
000002E0  43 69 73 63 6F 53 79 73 74 65 6D 73 07 00 01 00  CiscoSystems....
000002F0  08 00 01 01 09 00 01 00 0A 00 01 01 0B 01 00 99  ...............Ö
00000300  68 36 76 93 DD 4C C4 82 1F 34 49 60 88 36 99 05  h6vô L—é.4I`ê6Ö.
```

Based on the visible structure of this metadata region, it appears that the metadata header ends approximately around offset 0x000002F0. After this point, the data transitions into non-readable binary content.

```
cat9k-rpbase.16.12.0...
00000DE0  DB 2F 62 F3 AD 87 42 02 00 F1 BA 91 CF 8E 3E 8B  /b≤¡çB..±╣æ╧Ä>ï
00000DF0  35 25 BE EE 43 BA EB 45 D5 BB 23 DB 1C 82 8B 5A  5%╝εC╢δE╗#╛.éïZ
00000E00  D3 F2 0B FE 46 45 3D DE 2B F6 E5 E0 91 81 44 A6  ╙≥.▪FE=▐+÷σσæüDª
00000E10  6F EA 6C 4D B2 3D D4 4B 4E EE 80 08 5B A1 F1 ED  oΩlM▒=╘KNεÇ.[í±φ
00000E20  CA 3D 6A E5 9E 27 13 B0 EC 21 94 CB 51 D7 34 55  ╩═jσÞ'.╲∞!ö╦Q╫4U
00000E30  5E C1 E0 F6 56 17 89 98 07 21 CF 51 63 FD CC EB  ^┴αö÷V.ëÿ.!╧Qc²╠δ
00000E40  62 65 8D 21 46 25 77 49 74 FF BA 2A EF 13 5C 3F  beì!F%wIt ╢*∩.\?
00000E50  FF FA 8D CE 21 AB F6 71 84 17 AB 44 13 D4 5D F5   ·ì╬!½÷qä.½D.╘]⌡
00000E60  E0 12 12 10 42 22 E4 9E DE 62 13 67 8C C3 9A 63  α...B"ΣP▐b.gî├Üc
00000E70  DA 6A 49 EF 8A 30 06 98 63 55 B2 9C 2C 8E 6D 08  ┌jI∩è0.ÿcU▒£,Äm.
00000E80  32 99 D9 DE 96 90 85 4D 60 4F 3A B0 15 2D D8 8B  2Ö┘ ▐ûÉàM`O:╲.-╪ï
00000E90  4D DD 66 21 43 4C 65 68 C5 61 75 AA 81 4B 2C 61  M▌f!CLeh╪au¬üK,a
00000EA0  99 74 0E 25 ED 61 FC 32 A9 35 1F B2 C2 9B 92 69  Öt.%φaⁿ2⌐5.▒┬¢Æi
00000EB0  81 DF 50 AF A0 4D 63 5C 5C 21 6E 11 E8 DF DA A7  ü▀P»áMc\\!n.Φ▀┌°
00000EC0  B5 7B 01 C2 A4 E4 1F 3C 26 1A EA 44 9B 6A E6 B8  ╡{.┬ñΣ.<&.ΩD¢µ╕
00000ED0  48 7E A6 95 75 40 8C 83 DD 62 28 60 DC AF F5 73  H~ªòu@îâ▌b(`▄»⌡s
00000EE0  3E C0 9E 62 D8 8D EA 4B A9 F8 53 81 8A 48 4E 1B  >└Þb╪ìΩK⌐°SüèHN.
00000EF0  BE 28 1D B3 56 39 82 2B 51 6F C5 15 33 7C 8F 3B  ╛(.│V9é+Qo┼.3|Å;
00000F00  8E 0E 34 54 0D 55 E6 4B DF D4 A3 4D 0C E5 4A 41  Ä.4T.UµK▀╘úM.σJA
00000F10  99 12 B4 68 0B 3D 4D 30 BC F6 B4 F1 96 17 DB 41  Ö.┤h.=M0╝÷┤±û.▐A
00000F20  56 B5 2C 4B FF EE 56 E4 99 15 10 02 E3 93 86 5F  V╡,K εVΣÖ...πôå_
00000F30  93 A3 0C 0E 62 7B 11 E1 C3 4B C4 BE 17 73 19 7E  ôú..b{.ß├K─╛.s.~
00000F40  3A 52 28 5D 45 59 04 B3 F6 8E A9 0A 39 A5 E5 8D  :R(]EY.│÷Ä⌐.9Ñσì
00000F50  72 AD 9F 44 62 B5 A5 16 0B 0F 00 5D E8 F8 F7 42  r¡ƒDb╡Ñ....]Φ°≈B
00000F60  8E 9A 21 4F 97 9C 60 86 CE 86 E5 CE 42 64 34 70  ÄÜ!Oùε`å╬åσ╬Bd4p
00000F70  A3 00 9B C9 28 A9 E1 EC 17 0C 37 9C AA 86 E5 A2  ú.¢╒(⌐ß∞..7£¬åσó
00000F80  65 02 C6 CE AC 7B 67 C2 D2 77 67 2D 6A 14 8E B9  e.╠╬¼{g╥wg-j.Ä╣
00000F90  B0 2E FC 3E AF 20 53 86 41 60 F1 9D B1 A5 78 6F  ╲.ⁿ>» SåA`±¥╲Ñxo
00000FA0  F8 C3 BF D1 C0 B5 1F 8A 95 D0 80 7D 7C FC B1 F6  °├┐╤└╡.èò╨Ç}|ⁿ▒÷
00000FB0  31 3D 81 60 68 15 B9 35 AD C8 59 54 E4 F4 52 0A  1=ü`h.╢5¡╚YTΣ⌠R.
00000FC0  63 4A 25 9F 04 A3 02 7D 0C 2C BE FD 17 3C E7 FE  cJ%ƒ.ú.}.,╛²².<τ■
00000FD0  A7 23 BF 04 C9 86 A7 AF D4 CC 32 29 49 DA 4D 66  º#┐.╒åº»╘╠2)I┌Mf
00000FE0  CE FC 40 49 36 7A 7F CD 20 1B B2 E5 AC 3C 26 73  ╬ⁿ@I6z⌂═ .▒σ¼<&s
00000FF0  67 42 B6 3E 95 F3 78 D3 F2 A6 B8 22 F6 6B 45 A1  gB╢>òx≤x╙≥ªª"÷kEí
00001000  EC 31 B4 05 5E 5B C4 CA 4F 72 CF 23 F9 ED 8C 48  ∞1┤.^[─╩Or═#·φîH
00001010  20 E7 2C 11 41 93 32 E0 39 8C 66 72 9B ED AD C1   τ,.Aô2α9îfr¢φ¡┴
```



cat9k-rpbase.16.12.0…

```
00001DE0  D9 A8 3B 16 23 8B D0 C8 97 D8 6D 15 0A 58 9C 63  ┘¿;.#ï╩╚ù╪m..X£c
00001DF0  B7 DB 07 4F 21 48 ED E1 BB AB D6 1E 04 81 1A DE  ╗▌.O!Hφß╕½╓..ü.▐
00001E00  ED E3 38 E2 01 DA 0D F6 87 CF 7B 76 5F B9 96 F7  φπ8Γ.┌.÷ç╧{v_╡╢û≈
00001E10  85 83 A6 47 6A 7E 5C 50 DC 91 7B 77 08 D7 4E 56  àâªGj~\P▄æ{w.╫NV
00001E20  31 AD 70 36 49 D8 7F 6B 23 AB 0B 07 6E 71 47 21  1¡p6I╪⌂k#½..nqG!
00001E30  5A D9 0C BA 88 25 C6 25 7B 70 8D 10 C3 BA D7 19  Z┘.╢êº%╞%{pì.├╢╫.
00001E40  14 94 D9 C0 A4 24 43 80 A6 D4 C1 23 2D E8 81 73  .ö┘└ñ$CÇª╙┴#-Φüs
00001E50  35 31 29 70 36 1E 23 35 5E 83 AB 19 5E 4C C3 88  51)p6.#5^â½.^L├ê
00001E60  47 0F 6D E0 D2 73 BD 93 8B 66 45 97 05 F0 D5 82  G.mα╥s╝ôïfEù.≡╒é
00001E70  8E 8F 10 85 26 07 39 82 0C F6 CD B5 B5 EA 56 C7  ÄÅ.à&.9é.÷═╡╡ΩV╟
00001E80  CF 1C 8E A0 9F DD 6D A0 D2 97 85 D8 9B C7 00 1B  ╧.Ähƒ▌ má╥ùà╪¢╟..
00001E90  CE BF 92 F6 66 AA CC F1 B6 B7 71 B0 1B E5 18 C3  ╬╗Æ÷f¬╠ñ╢╖q╲.σ.├
00001EA0  FE 37 82 1C 3B 6E 9D 9E 4D FC 24 3A C5 C3 04 C4  ▪7é.;n¥₧Mⁿ$:┼├.─
00001EB0  F2 A5 65 08 DF 48 B4 DB 51 52 32 6E E6 60 4C AC  ≥Ñe.▀H┤╛QR2nµ`L¼
00001EC0  4E D7 65 3D 6D F5 B4 BA 32 5B 54 A0 EF 83 A8 62  N╫e=m⌡┐╢2[Tá∩â¿b
00001ED0  F4 E5 57 07 6B A5 06 A6 6D D9 05 04 DB A1 8C EB  ⌠σW.kÑ.ªm┘..▄í╬δ
00001EE0  3B 5A F8 59 5A 2A 7A CD 9B B5 49 59 6F 07 0C 5A  ;Z°YZ*z═¢╡IYo..Z
00001EF0  6B F1 FB 5D 6B 25 48 6B 23 2E DB 79 73 32 3D 79  k±√]k%Hk#.▄ys2=y
00001F00  6C ED 1D 81 AE EA 20 91 F7 8F E5 0B 01 BB 64 45  lφ.ü«Ω æ≈Åσ..╗dE
00001F10  07 E0 61 65 6D B2 76 DB E5 1D CE 95 6F 52 EF 84  .αaem▒v▄σ.╬òoR∩ä
00001F20  6E B6 86 7F 11 4A B7 66 CC B3 BC D6 2A FB DE FA  n╢å⌂.J╗f╠│╝╓*√▐·
00001F30  77 B9 07 44 59 76 A9 8B ED F9 58 C1 36 23 6F CD  w╣.DYv⌐ï φ·X┴6#o═
00001F40  41 5A E6 AA 42 94 01 99 FE E1 26 37 64 65 5B 59  AZµ¬Bö.Ö▪ß&7de[Y
00001F50  B0 65 10 03 F6 BC 3B 2B DA F6 43 FD 6A 58 82 42  ╲e..÷╝;+┌÷C²jXéB
00001F60  B7 54 F9 7D 4B 09 69 AC EA D9 87 B4 AA 15 E7 15  ╗T·}K.i¼½┘ç┤¬.τ.
00001F70  50 D8 F8 12 4B FF CF 23 3E 00 DF 32 8A 39 96 39  P╪°.K ╧#>.▀2è9û9
00001F80  AA DA 79 C6 A7 35 CD 61 50 58 92 50 6F 4F CF 7F  ¬┌y╞º5═aPXÆPoO╧⌂
00001F90  A7 16 6F 74 EC 8B 20 DC B0 B1 6D E5 90 9E A9 4D  º.ot∞ï ▄╲╲╲mσÉ₧⌐M
00001FA0  F0 C3 15 0F 12 5C 7B E1 30 98 5E 28 02 EF ED 58  ≡├...\{ßθÿ^(.∩φX
00001FB0  F7 F7 C9 D2 0E 51 FD B7 10 46 5A C1 34 6C EC 9E  ≈≈╔╥.Q²╗.FZ┴4l∞₧
00001FC0  2B 17 7E EA AF EF F9 20 0A EB 95 35 99 E2 4A 6E  +.~Ω»∩· .δò5ÖΓJn
00001FD0  3A 6A 57 D6 8B 54 49 68 85 90 12 67 EA BE C1 A2  :jW╥ïTIhàÉ.gΩ╜┴╩ó
00001FE0  05 91 44 F6 EC 97 1E 11 41 77 A1 4B 44 CD 65 3F  .æD÷∞ù..AwíKD═e?
00001FF0  EB AB 2A D3 72 DC 5B AC FF BA 35 18 74 53 57 3A  δ½*╙r▄[¼ ╟5.tSW:
00002000  94 B8 1E F8 42 BE 94 29 E4 8A 71 7B 98 7F 95 6E  ö╕.°B╛ö)Σèq{ÿ⌂òn
00002010  F4 A8 B0 6C EB FB 9A 26 85 B5 3C 4F 5B E6 B6 86  ⌠¿╲lδ√Ü&à╡<O⌠µ╢å
```

From roughly 0x000002F0 up to the inspected region around 0x00001000–0x00002000, the file mainly contains high-entropy binary data, which likely corresponds to the beginning of the binary payload section of the .pkg container.

Possible length fields

cat9k-rpbase.16.12.0...

```
000000D0  00 00 00 18 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000000E0  41 47 45 5F 42 4F 4F 54 41 52 43 48 00 00 00 09  AGE_BOOTARCH....
000000F0  00 00 00 0E 41 52 43 48 5F 69 36 38 36 5F 54 59  ....ARCH_i686_TY
00000100  50 45 00 00 00 00 00 09 00 00 00 14 4B 45 59 5F  PE..........KEY_
00000110  54 4C 56 5F 42 4F 41 52 44 5F 43 4F 4D 50 41 54  TLV_BOARD_COMPAT
00000120  00 00 00 09 00 00 00 10 42 4F 41 52 44 5F 63 61  ........BOARD_ca
00000130  74 39 6B 5F 54 59 50 45 00 00 00 09 00 00 00 18  t9k_TYPE........
00000140  4B 45 59 5F 54 4C 56 5F 43 52 59 50 54 4F 5F 4B  KEY_TLV_CRYPTO_K
00000150  45 59 53 54 52 49 4E 47 00 00 00 09 00 00 00 04  EYSTRING........
00000160  6E 6F 6E 65 00 00 00 09 00 00 00 0B 43 57 5F 42  none........CW_B
00000170  45 47 49 4E 3D 24 24 00 00 00 00 09 00 00 00 11  EGIN=$$.........
00000180  43 57 5F 46 41 4D 49 4C 59 3D 24 63 61 74 39 6B  CW_FAMILY=$cat9k
00000190  24 00 00 00 00 00 00 09 00 00 00 28 43 57 5F 49  $..........(CW_I
000001A0  4D 41 47 45 3D 24 63 61 74 39 6B 2D 72 70 62 61  MAGE=$cat9k-rpba
000001B0  73 65 2E 31 36 2E 31 32 2E 30 37 2E 53 50 41 2E  se.16.12.07.SPA.
000001C0  70 6B 67 24 00 00 00 09 00 00 00 15 43 57 5F 56  pkg$........CW_V
000001D0  45 52 53 49 4F 4E 3D 24 31 36 2E 31 32 2E 30 37  ERSION=$16.12.07
000001E0  24 00 00 00 00 00 00 09 00 00 00 37 43 57 5F 46  $..........7CW_F
000001F0  55 4C 4C 5F 56 45 52 53 49 4F 4E 3D 24 31 36 2E  ULL_VERSION=$16.
00000200  31 32 2E 30 37 2E 30 2E 36 35 36 35 2E 31 36 34  12.07.0.6565.164
00000210  33 38 32 39 39 39 36 2E 2E 47 69 62 72 61 6C 74  3829996..Gibralt
00000220  61 72 24 00 00 00 00 09 00 00 00 34 43 57 5F 44  ar$........4CW_D
00000230  45 53 43 52 49 50 54 49 4F 4E 3D 24 43 69 73 63  ESCRIPTION=$Cisc
00000240  6F 20 49 4F 53 20 53 6F 66 74 77 61 72 65 2C 20  o IOS Software,
00000250  49 4F 53 2D 58 45 20 53 6F 66 74 77 61 72 65 24  IOS-XE Software$
00000260  00 00 00 09 00 00 00 09 43 57 5F 45 4E 44 3D 24  ........CW_END=$
00000270  24 00 00 00 00 00 00 0C 00 00 01 88 01 00 02 01  $..........ê....
00000280  01 02 00 04 00 00 01 7C 03 00 61 04 00 28 43 4E  .......|..a..(CN
00000290  3D 43 69 73 63 6F 53 79 73 74 65 6D 73 3B 4F 55  =CiscoSystems;OU
000002A0  3D 49 4F 53 2D 58 45 3B 4F 3D 43 69 73 63 6F 53  =IOS-XE;O=CiscoS
000002B0  79 73 74 65 6D 73 05 00 08 36 31 46 41 45 35 36  ystems...61FAE56
000002C0  30 06 00 28 43 4E 3D 43 69 73 63 6F 53 79 73 74  0..(CN=CiscoSyst
000002D0  65 6D 73 3B 4F 55 3D 49 4F 53 2D 58 45 3B 4F 3D  ems;OU=IOS-XE;O=
000002E0  43 69 73 63 6F 53 79 73 74 65 6D 73 07 00 01 00  CiscoSystems....
000002F0  08 00 01 01 09 00 01 00 0A 00 01 01 0B 01 00 99  ................Ö
00000300  68 36 76 93 DD 4C C4 82 1F 34 49 60 88 36 99 05  h6vô▌L—é.4I`ê6Ö.
```

Several readable metadata fields were identified, including CW_IMAGE, CW_VERSION, CW_DESCRIPTION, and CW_END.

Each of these entries is preceded by a 4-byte value such as 00 00 00 28, 00 00 00 15, 00 00 00 34, and 00 00 00 09. These values appear to represent the length of the following metadata string.

Because this pattern repeats for multiple fields, it suggests that the metadata header is organized in a TLV-like structure, where a length field is followed by a textual metadata value. This repeating structure indicates that the beginning of the .pkg container contains a structured TLV metadata block that stores package attributes such as the image name, version, and description before the binary payload begins.

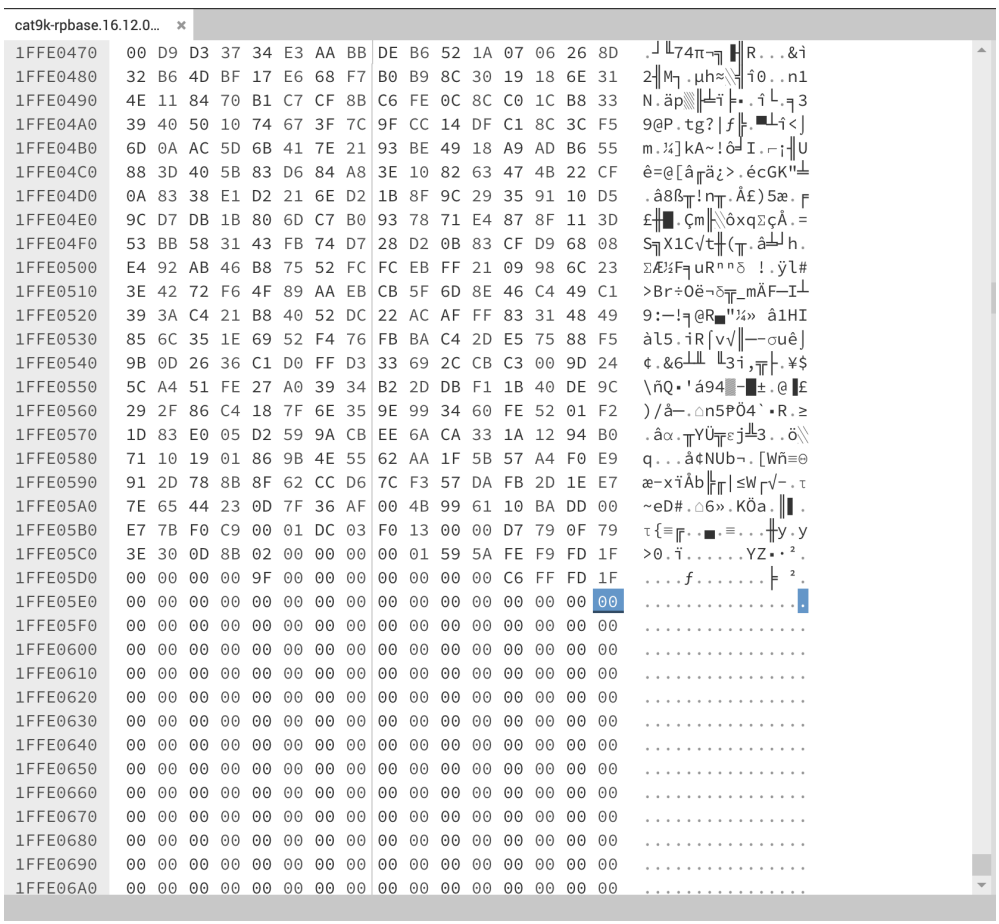
Repeating patterns

Repeating structural pattern becomes visible. Each metadata entry appears to follow the same layout: a 4-byte value followed by a readable ASCII metadata string.

This pattern repeats for multiple entries in sequence, including fields such as CW_IMAGE, CW_VERSION, CW_DESCRIPTION, and CW_END.

The repeated layout of [4-byte value] → [metadata string] strongly indicates that the header is organized as a structured metadata block.

Payload Region Observation



To better understand the internal structure of the .pkg container, the file was inspected not only at the beginning but also near the end of the package using a hex editor.

After the metadata header region (ending approximately around offset 0x000002F0), the file content changes to non-readable binary data. This pattern continues throughout the remainder of the file.
Inspection near the end of the file confirms that the package is almost entirely composed of this binary data region. No additional readable metadata structures similar to the header were observed further in the file.
This observation supports the assumption that the .pkg container consists of a small structured metadata header followed by a large binary payload region containing the actual runtime components of the firmware package.

.pkg file structure

.pkg container layout (observed)

```
0x00000000 ──────────────────────────────
Metadata Header (TLV-like)
  CW_BEGIN
  CW_IMAGE
  CW_VERSION
  CW_DESCRIPTION
  CW_END

~0x000002F0 ─────────────────────────────
Transition from metadata to payload

0x000002F0+ ─────────────────────────────
Binary Payload Region
(compressed modules)
```